

Predictive Visual Learning Builds a Visual Prototype Bias: Evidence from I-JEPA Representational Geometry

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Abstract

We investigate whether I-JEPA, a vision model trained by predicting masked patch representations in latent space, recapitulates human prototype structure across 1,854 object concepts (THINGS dataset). Using RSA against human similarity judgements, typicality norm validation, layer-by-layer probing across all 32 transformer layers, and 4.58 million odd-one-out trials, we identify a systematic dissociation: I-JEPA typicality correlates positively with human norms for perceptually-defined categories (birds, fruit, sea animals; $r = +0.275$) but negatively for functionally-defined categories (vehicles, tools, weapons; $r = -0.266$), yielding $\Delta r = +0.541$ ($p < 0.001$). This dissociation is significant only in I-JEPA; DINOv2 shows a weaker non-significant trend ($\Delta r = +0.141$, $p = 0.076$). The dissociation grows with depth, accelerating in the final 12 layers where RSA alignment peaks then degrades. Attention maps reveal late-layer collapse onto a single dominant patch, providing a mechanistic account. I-JEPA's visual prediction objective is sufficient for perceptual prototype structure but insufficient for conceptual prototype structure.

Keywords: representational similarity analysis; prototype theory; self-supervised learning; visual cognition; THINGS

Introduction

Rosch's prototype theory (Rosch, 1975) holds that category typicality reflects proximity to a category's central tendency. For functionally-defined categories (vehicles, tools), this central tendency is governed by function rather than visual appearance: a car is more typical than an airboat because it better fulfils the vehicle role, not because it is more visually average. Self-supervised vision models are increasingly used as proxies for human visual representation (Yamins & DiCarlo, 2016; Kornblith et al., 2019). I-JEPA (Assran et al., 2023) trains a ViT-H/14 to predict latent representations of masked image patches from unmasked context—unlike MAE, which predicts pixels, I-JEPA operates entirely in representation space, making it a natural analogue to Rao and Ballard's (1999) predictive coding model. We examine whether I-JEPA's representational geometry recapitulates human prototype structure across 1,854 THINGS concepts.

Methods

Dataset. The THINGS dataset (Hebart et al., 2019) provides 1,854 object concepts (~12 images each). Human similarity was obtained from the SPoSE 66-dimensional embedding

(Hebart et al., 2020). Typicality norms came from THINGSplus (Stoinski et al., 2023), providing human typicality ratings across 53 categories.

Models. I-JEPA ViT-H/14 was compared against DINOv2 ViT-L/14, CLIP ViT-L/14, DINO ViT-B/16, and MAE ViT-L/16. Concept embeddings were extracted by mean-pooling final-layer patch tokens (~12 images per concept) and L2-normalising. Layer-by-layer representations used forward hooks on a single image per concept (accounting for minor discrepancies with the main results).

Typicality dissociation. Model typicality was operationalised as negative cosine distance to the category centroid. Spearman r was computed against human norms. Of 53 THINGSplus categories, 27 were classified as *aligned* (visual $\approx$ conceptual centrality; 13 categories) or *misaligned* (function determines typicality; 14 categories); 26 were excluded due to ambiguous classification. Significance was assessed via label permutation (10,000 iterations).

Layer-by-layer probing. RSA and typicality correlations were computed for each of 32 transformer layers. Attention entropy and top-10% patch concentration were measured at layers 4, 20, and 28 for three exemplars (rollerskate, airboat, wagon).

Triplet validation. Model agreement with human odd-one-out choices was computed across 4.58 million THINGS triplets, partitioned into within-aligned, within-misaligned, and cross-boundary types.

Results

RSA and model comparison. I-JEPA achieved Spearman $r = 0.386$ ($p < 0.001$), ranking second (Table 1). DINOv2 ranked highest ($r = 0.489$); MAE was near chance ($r = 0.026$), confirming training objective as the primary determinant of human alignment.

Table 1. Model comparison. Typicality $\Delta r = \text{aligned minus misaligned mean } r$. $\dagger p = 0.076$; $*** p < 0.001$; $ns = \text{not significant}$.

Model	RSA r	Trip. Δ	Typ. Δr
I-JEPA ViT-H/14	.386	+6.3%	+0.541***
DINOv2 ViT-L/14	.489	+15.4%	+0.141 $\dagger$
CLIP ViT-L/14	.271	+13.3%	−0.006 ns
DINO ViT-B/16	.364	+7.6%	−0.015 ns
MAE ViT-L/16	.026	+0.7%ns	+0.008 ns

Typicality dissociation. I-JEPA showed a strong dissociation ($\Delta r = +0.541$, $p < 0.001$), the only model to reach significance. DINOv2 showed a directionally similar but non-significant trend ($\Delta r = +0.141$, $p = 0.076$); all other baselines showed near-zero Δr (range: -0.015 to $+0.008$, all ns). I-JEPA captures visual averageness as its prototype signal—matching human typicality for perceptual categories but actively inverting it for functional ones.

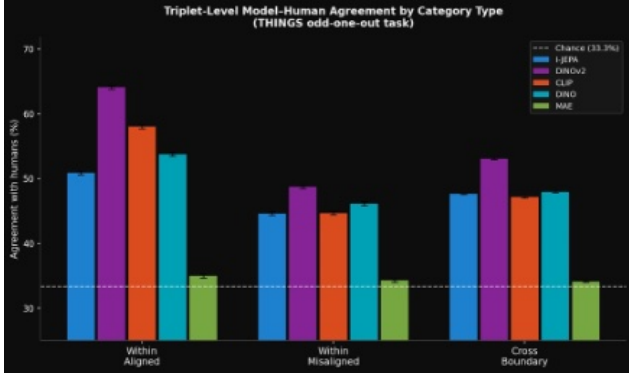


Figure 1. Layer-by-layer typicality dissociation. *Top*: aligned correlations increase with depth (left); misaligned correlations invert in final layers (right). *Bottom*: Δr grows with depth, significant from layer 16 (shaded); RSA peaks at layer 20 then declines (right).

Layer-by-layer analysis. RSA increased from $r = 0.19$ (layer 0) to $r = 0.41$ (layer 20), then degraded ($r = 0.31$ at layer 31). The typicality Δr was -0.09 at layer 0, crossed zero near layer 3, reached significance at layer 16, and accelerated sharply after layer 20 (peak $\Delta r = 0.51$ at layer 31). Misaligned correlations inverted ($r = -0.24$ at layer 31) while aligned correlations continued to improve (Figure 1).

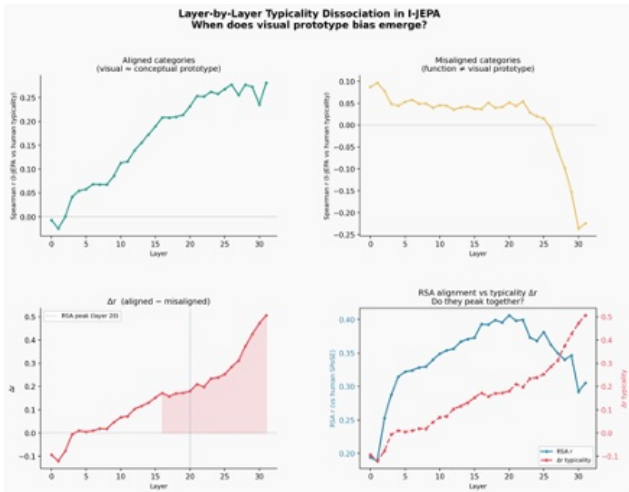


Figure 2. Attention maps at layers 4, 20, and 28. Layer 4: distributed spatial attention. Layer 20 (RSA peak): partial focusing. Layer 28: collapse onto a single dominant patch.

Attention analysis. Attention entropy decreased across sampled layers (4, 20, 28). By layer 28, top-10% patch concentration rose from 0.187 to 0.511 for rollerskate

(Figure 2). The predictive objective drives late-layer representations toward visually salient anchors, amplifying visual averageness at the cost of semantic organisation.

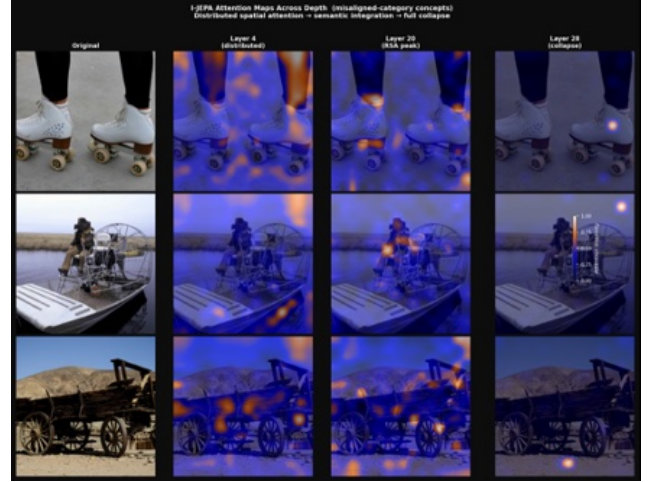


Figure 3. Triplet-level model–human agreement by category type (4.58M trials). I-JEPA shows the smallest within-aligned advantage of any semantically capable model.

Triplet validation. I-JEPA showed the smallest aligned/misaligned agreement gap (+6.3%) of any semantically capable model (Figure 3). DINOv2 (+15.4%) and CLIP (+13.3%) showed substantially larger gaps, consistent with stronger categorical organisation. MAE (+0.7%, ns) showed no reliable gap.

Discussion

I-JEPA's visual prediction objective produces a representational geometry with a specific failure mode: prototype structure matching human typicality where visual and conceptual centrality co-align, but systematically inverting for functionally-defined categories. This dissociation is significant only in I-JEPA; DINOv2's trend is four-fold smaller ($p = 0.076$), implicating the predictive SSL objective as a key factor.

The layer analysis shows that the dissociation emerges progressively and accelerates in the final 12 layers, where semantic alignment peaks then degrades. Attention collapse in these layers provides a mechanistic link: late-layer attention converges on visually salient anchors, strengthening visual averageness at the cost of broader semantic organisation.

These findings carry practical implications for computational cognitive modelling. I-JEPA is a reliable proxy for human perceptual similarity but systematically misleading for conceptual category structure. Conceptual prototype structure—particularly for functional categories—requires learning signals beyond visual self-prediction, such as language supervision, social context, or embodied interaction.

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